

Ergodicity (Boltonmann.
 ergon "energy, work"
 hodos "way, path")

(1)

①
 $(X, \mathcal{B}, \mu) \supset f$ μ probability measure

Def. f is ergodic if $\forall A \in \mathcal{B}, f^{-1}(A) = A$.

$$\Rightarrow \mu(A) = 1 \text{ or } 0.$$

(Stronger version. $\mu(f^{-1}A \Delta A) = 0 \Rightarrow \mu(A) = 1 \text{ or } 0$)

$\varphi: X \rightarrow \mathbb{C}$ measurable. — φ is f -invariant

$$\text{if } \varphi \circ f = \varphi.$$

— φ is a.e. f -invariant if $\varphi \circ f = \varphi$ a.e. μ

Prop. TFAE (The following are equivalent)

1. f ergodic
2. $\forall \varphi$ ^{measurable} a.e. f -invariant $\Rightarrow \varphi$ a.e. const. function.
3. $\forall \varphi \in L^1(X, \mu)$ a.e. f -inv $\Rightarrow \varphi = \text{constant}$ a.e.
4. $\forall \varphi \in L^2(X, \mu)$ a.e. f -inv $\Rightarrow \varphi = \text{constant}$ a.e.

pf. $L^2 \subset L^1$, (4) $\Rightarrow$ 3)

$$3) \Rightarrow 2), \quad \varphi \circ f = \varphi \text{ a.e.} \quad \varphi_N = \min\{\varphi, N\} \in L^1.$$

$$\stackrel{3')}{\Rightarrow} \varphi_N \circ f = \varphi \text{ a.e.} \Rightarrow \varphi_N = \text{const a.e.}$$

$$2) \Rightarrow 1). \quad f^{-1}(A) = A, \quad \varphi = 1_A \Rightarrow \varphi \circ f = \varphi \stackrel{2')}{\Rightarrow} \varphi = \text{const. a.e.}$$

$$1) \Rightarrow 4) \quad A = \{x \mid a \leq \varphi(x) < b\}$$

$$\varphi \circ f = \varphi \quad \text{a.e. } \mu. \quad \Rightarrow \quad \mu(f^{-1}(A) \Delta A) = 0 \quad (2)$$

$$B = \bigcup_{k \geq 0} f^{-k}(A). \quad f^{-1}(B) \subset B, \quad \mu(f^{-1}(B)) = \mu(B) = \mu(A)$$

$$\mu(B) = \mu(A).$$

$$C = \bigcap_{\ell \geq 0} f^{-\ell}(B). \quad f^{-1}(C) = C, \quad \mu(C) = \mu(B) \quad \square$$

Prop: X compact metric space, μ probability, $\text{supp } \mu = X$,

f preserves μ , continuous. The μ -a.e. x

$$\overline{\{f^n(x), n \in \mathbb{N}\}} = X$$

pf: X compact metric space $\Rightarrow \exists \mathcal{C}$ countable family of open sets, top base. $\forall U_k \in \mathcal{C}$.

$$A_k = \bigcup_{n \geq 0} f^{-n}(U_k) \quad (x, \exists n \text{ s.t. } f^n(x) \in U_k)$$

$$f^{-1}(A_k) \subset A_k \quad \xRightarrow{\text{stronger version}} \quad \mu(A_k) = 0 \text{ or } 1$$

$$\text{Due to } \text{supp } \mu = X. \quad \Rightarrow \quad \mu(U_k) > 0, \quad \mu(A_k) = 1$$

$$A = \bigcap_{k \geq 1} A_k$$

$$\mu(A) = 1$$

$$\text{Q. } \forall x \in A. \forall k, x \in A_k, \exists n. \text{ s.t. } f^n(x) \in U_k \quad \square$$

$$\text{Ex. } \mathbb{T}^N = (\mathbb{R}/\mathbb{Z})^N, \quad \mu = \text{Leb.}, \quad B$$

$$i). \quad R_\alpha(x) = (x_1 + \alpha_1, \dots, x_N + \alpha_N), \quad \text{preserve Leb}$$

$$R_\alpha \text{ ergodic} \Leftrightarrow 1, \alpha_1, \dots, \alpha_N \text{ rational independent}$$

ii) M integer ≥ 2 . $g: \theta \rightarrow M\theta$ ergodic (3)

Rk. What about a matrix $A \in M_{M \times N}(\mathbb{Z})$. $\theta \rightarrow A\theta$.

Pf. Use Fourier coefficients. $\widehat{\varphi}(k) = \int_{\mathbb{T}^N} \varphi e^{2\pi i k \cdot \theta} d\text{Leb}(\theta)$
 $k \in \mathbb{Z}^N$,
 ~~$\alpha_1, \dots, \alpha_N$ rational dependent.~~

$$\begin{aligned}\widehat{\varphi \circ R_\alpha}(k) &= \int \varphi \circ R_\alpha e^{2\pi i k \cdot \theta} d\theta \\ &= \int \varphi e^{2\pi i k \cdot (R_\alpha^{-1}\theta)} d\theta \\ &= e^{2\pi i (k \cdot (-\alpha))} \widehat{\varphi}(k)\end{aligned}$$

$$\widehat{\varphi \circ g}(k) = \int \varphi \circ g(\theta) e^{2\pi i k \cdot \theta} d\theta$$

$2\pi M | k$.

$$= \widehat{\varphi}(k/M)$$

i) If φ is a.e. f -inv. $\Rightarrow \forall k$

$$\widehat{\varphi}(k) = \widehat{\varphi \circ R_\alpha}(k) = e^{2\pi i (k \cdot (-\alpha))} \widehat{\varphi}(k)$$

① $(1, \alpha)$ rational dependent, $\exists k. k \cdot \alpha \in \mathbb{Z}$.

$$\widehat{\varphi}(k) = 1, \leadsto \varphi: \theta \mapsto e^{-2\pi i (k \cdot \theta)}, f\text{-inv}$$

② $(1, \alpha)$ rational independent. $\forall k \neq 0, k \cdot \alpha \notin \mathbb{Z}$

$$\Rightarrow \widehat{\varphi}(k) = 0. \quad \forall k \neq 0. \quad \Rightarrow \varphi = \text{const a.e.}$$

ii) $\widehat{\varphi}(k) = \widehat{\varphi}(k \cdot M) = \widehat{\varphi}(k \cdot M^2) \dots$

$$\varphi \in L^2, \quad \widehat{\varphi} \in L^2 \Rightarrow \widehat{\varphi}(k) = 0. \quad \forall k \neq 0$$

Ex 2. (Λ^N, σ) see exercise.

(4)

$\sigma^{-1}(E) = \sigma^{-1}(E) \cap E, \exists A$ finite union of cylinders. $\mu(E \Delta A) < \epsilon$. $\mu(E) = \mu(\sigma^{-n}(E) \cap E)$
 Ergodic theorem $\exists n, \mu(\sigma^{-n}(A) \cap A) = \mu(A)^2, \Rightarrow \mu(E) = \mu(E)^2$

Thm. (Birkhoff). $(X, \mathcal{B}, \mu)$. μ ~~prob~~, f .

μ is f -inv

$\forall \varphi \in L^1(X, \mu)$. Write

$$S_n \varphi(x) = \frac{1}{n} \sum_{j=0}^{n-1} \varphi \circ f^j(x)$$

(i). The limit $\hat{\varphi}(x) = \lim_{n \rightarrow \infty} S_n \varphi(x)$ exists a.e. μ

(ii) $\hat{\varphi} \circ f(x) = \hat{\varphi}(x)$ a.e. μ

(iii) $\|\hat{\varphi}\|_{L^1} \leq \|\varphi\|_{L^1}$

(iv). μ finite. L^1 -convergence

$$\|S_n \varphi - \hat{\varphi}\|_{L^1} \rightarrow 0$$

(v) $\forall A \in \mathcal{B}, \mu(A) < \infty, f^{-1}(A) = A$ then

$$\int_A \varphi d\mu = \int_A \hat{\varphi} d\mu$$

(vi) If μ ergodic probability, then

$$\hat{\varphi}(x) = \int \varphi d\mu \quad \text{a.e.}$$

$S_n \varphi$ called Birkhoff average.

$\lim S_n \varphi$ called time average, orbit average

$\int \varphi d\mu$ space average

Cor. $A \in \mathcal{B}$, $\varphi = 1_A$

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$$S_n \varphi(x) = \frac{1}{n} \# \{ j \leq n \mid f^j(x) \in A \}$$

μ prob., ergodic $S_n \varphi(x) \rightarrow \mu(A)$ a.e.

Cor. a.e. $x \in [0, 1[$ $\forall j \in \{0, 1, \dots, 9\}$

$x = 0.x_1x_2\dots$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \# \{ k \leq n \mid x_k = j \} = 1/10$$

pf. $[0, 1[\simeq \mathbb{R}/\mathbb{Z}$. $\forall j \rightarrow I_j \subset \mathbb{R}/\mathbb{Z}$, $\mu(I_j) = 1/10$

$f: x \rightarrow 10x$. $\mu = \text{Leb.}$ - f -ergodic $\square$
 $\{k \leq n \mid f^k(x) \in I_j\} = \{k \leq n \mid x_k = j\}$

pf. ergodic thm.

Lem. (Maximal ergodic thm)

$\varphi \in L^1(X, \mu, \mathbb{R})$

$$\varphi^*(x) = \max_{k \in \mathbb{Z}} \sup_{n \geq 1} S_n \varphi(x)$$

then

$$\int_{\{\varphi^*(x) > 0\}} \varphi d\mu \geq 0$$

pf. (Tricky). Let $\psi_n = \max \{0, \varphi, \varphi \circ f + \varphi, \dots, \varphi \circ f^{n-1} + \varphi\}$

On $\{\psi_n > 0\}$, we have

$$\psi_{n-1} \circ f + \varphi \geq \max_{0 \leq k \leq n-1} \{ \varphi \circ f^k + \dots + \varphi \}$$

$$\text{due to } \psi_n(x) > 0 \geq \max_{-1 \leq k \leq n} \{ \varphi \circ f^k + \dots + \varphi \} = \psi_n$$

Therefore

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$$\begin{aligned} \int_{\{\psi_n > 0\}} \varphi &\geq \int_{\{\psi_n > 0\}} \psi_n - \int_{\{\psi_n > 0\}} \psi_{n-1} \circ f \\ &= \int_X \psi_n - \int_{\{\psi_n > 0\}} \psi_{n-1} \circ f \\ (\text{due to } \psi_{n-1} \circ f \geq 0) &\geq \int_X \psi_n - \int_X \psi_{n-1} \circ f = \int_X \psi_n - \psi_{n-1} \circ f \geq 0 \\ &\quad \psi_n \geq \psi_{n-1} \circ f. \end{aligned}$$

$$\text{Now } \{\varphi^* > 0\} = \bigcup \{\psi_n > 0\}, \quad \{\psi_n > 0\} \subset \{\psi_{n+1} > 0\}$$

$$\Rightarrow \int_{\{\varphi^* > 0\}} \varphi \geq 0$$

(Back to Pf of Birkhoff ergodic thm), $\forall \alpha < \beta$

$$\text{Let } E_{\alpha, \beta} = \{x \mid \liminf S_n \varphi(x) < \alpha < \beta < \limsup S_n \varphi(x)\}$$

$$\text{Notice } T^{-1} E_{\alpha, \beta} = E_{\alpha, \beta}. \quad \text{Apply lem with } X = E_{\alpha, \beta}$$

$$\text{Due to } \{(\varphi - \beta)^* > 0\} \cap E_{\alpha, \beta} = E_{\alpha, \beta},$$

$$\int_{E_{\alpha, \beta}} \varphi - \beta = \int_{E_{\alpha, \beta} \cap \{(\varphi - \beta)^* > 0\}} \varphi - \beta \geq 0$$

$$\int_{E_{\alpha, \beta}} \varphi \geq \beta \mu(E_{\alpha, \beta})$$

$$\text{Consider } -\varphi + \alpha, \text{ similarly, } \int_{E_{\alpha, \beta}} \varphi \leq \alpha \mu(E_{\alpha, \beta})$$

$$\text{Hence, } \mu(E_{\alpha, \beta}) = 0. \quad \forall \alpha, \beta$$

$$\text{Therefore, a.e. } x, \quad \liminf S_n \varphi(x) = \limsup S_n \varphi(x) \quad \square$$

(ii) $\tilde{\varphi} \circ f = \lim S_n(\varphi \circ f)$ (7)

$$= \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} S_{n+1} \varphi - \frac{1}{n} \varphi \right) S_n \varphi = \tilde{\varphi} \text{ a.e.}$$

(iii) ~~$\forall N, \varphi_N = \min\{|\varphi|, N\}$~~
 Suppose $\varphi \geq 0$. then Fatou lem.

$$\int \tilde{\varphi} = \int \lim S_n \varphi \leq \liminf \int S_n \varphi = \int \varphi$$

(iv) Use dominated convergence. $\varphi_N = \begin{cases} \min\{\varphi, N\} & \varphi \geq 0 \\ \max\{\varphi, -N\} & \varphi < 0 \end{cases}$
 $\|\varphi - \varphi_N\|_1 \xrightarrow{\text{as } N \rightarrow \infty} 0$

$$\|S_n \varphi_N - \tilde{\varphi}_N\|_1 \xrightarrow{N \rightarrow \infty} 0 \text{ due dominated convergence}$$

$$\|S_n(\varphi - \varphi_N)\|_1 \leq \|\varphi - \varphi_N\|_1 \rightarrow 0$$

$$\|\tilde{\varphi}_N - \tilde{\varphi}\|_1 = \|(\varphi_N - \varphi)\|_1 \leq \|\varphi_N - \varphi\|_1 \rightarrow 0$$

(v) Due to $f^{-1}(A) = A$. $\int_A \varphi \circ f d\mu = \int_A \varphi d\mu$

Hence

$$\int_A \varphi d\mu = \int_A S_n \varphi d\mu \xrightarrow[n \rightarrow \infty]{(iv)} \int_A \tilde{\varphi} d\mu$$

(vi) (ii) + ergodic $\Rightarrow \tilde{\varphi} = \text{cst}$ a.e.

$\tilde{\varphi}(x) = \int \tilde{\varphi} = \lim \int S_n \varphi = \int \varphi$ □
 $(\Omega, \mathcal{A}, P)$ probability space, $(\mathbb{R}, \mathcal{B})$ random variable. $X_i: (\Omega, \mathcal{A}) \rightarrow (\mathbb{R}, \mathcal{B})$

Example: (law of large numbers) real

Let $X_1, X_2, X_3, \dots$ be a sequence of independent, identical distributed random variables with finite expectation.

Then a.e. $\frac{1}{n} (X_1 + X_2 + \dots + X_n) \rightarrow E(X_1)$

Add proof of ergodicity of $\mu^{\otimes \mathbb{N}}$ Here

Let μ be the distribution of X_1 , & $\mu = (X_1, \mathbb{P})$ (8)

$$\mu(A) = \mathbb{P}(X_1(\omega) \in A) \quad \forall A \in \mathcal{B} = \mathcal{B}(\mathbb{R})$$

Independent.

$$A_{k_i} \in \mathcal{B}$$

$$\mathbb{P}(X_{k_i} \in A_{k_i}, \text{ } k_i \text{ finite}) = \prod \mathbb{P}(X_{k_i} \in A_{k_i})$$

Pf. Construct a space. $(X = \mathbb{R}^{\mathbb{N}}, \mathcal{B}, \mu^{\otimes \mathbb{N}})$ σ shift

$$\varphi: X \rightarrow \mathbb{R}, \quad (x_1, x_2, \dots) \mapsto x_1$$

Distribution of $(X_1, X_2, X_3, \dots)$ (independence.)

identical to ~~$(x_1, x_2, \dots)$~~ with measure $\mu^{\otimes \mathbb{N}}$.

$$v: \Omega \rightarrow \mathbb{R}^{\mathbb{N}}, \quad v(\omega) = (X_1(\omega), X_2(\omega), \dots) \quad (v)_* \mathbb{P} = \mu^{\otimes \mathbb{N}}$$

$$\frac{1}{n} (X_1 + X_2 + \dots + X_n) = S_n \varphi (x_1^{(\omega)}, x_2^{(\omega)}, x_3^{(\omega)}, \dots) = S_n \varphi \circ v(\omega)$$

(Exercise, $\mu^{\otimes \mathbb{N}}$ is σ -ergodic.). By Birkhoff ergodic

thm,

$$\text{a.e. } S_n \varphi \rightarrow \tilde{\varphi} = \int x d\mu = \mathbb{E}(X_1)$$

$$(\Omega, \mathcal{A}, \mathbb{P}) \xrightarrow{\sigma} (\Omega, \mathcal{A}, \mathbb{P})$$

$$v: \downarrow \text{semi-conjugate} \quad \downarrow v$$

$$(\mathbb{R}^{\mathbb{N}}, \mu^{\otimes \mathbb{N}}) \xrightarrow{\sigma} (\mathbb{R}^{\mathbb{N}}, \mu^{\otimes \mathbb{N}})$$

$$K = \left\{ \omega \mid \lim_{n \rightarrow \infty} \frac{1}{n} (X_1(\omega) + \dots + X_n(\omega)) \rightarrow \mathbb{E}(X_1) \right\}$$

$$K' = \left\{ x \mid \lim_{n \rightarrow \infty} \frac{1}{n} (\varphi(x) + \varphi(\sigma x) + \dots + \varphi(\sigma^{n-1} x)) \rightarrow \mathbb{E}(X_1) \right\}$$

$$v^{-1}(K') = K$$

Thm von Neumann ergodic thm

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Let $(X, \mathcal{B}, \mu)$ probability space. $f \in \mathcal{G} X$ preserve μ .

Then $\forall \varphi \in L^2(X, \mu)$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} \varphi \circ f^k = P\varphi \quad \text{in } L^2. \quad (\star)$$

where P the projection from $L^2(X, \mu)$ to the subspace of f -inv functions.

Pf: Let $\mathcal{H} \subseteq L^2(X, \mu)$, f -inv subspace.
 $U\varphi = \varphi \circ f$.

then due to $f\mu$ is f -inv. $\|U\varphi\|_{L^2} = \|\varphi\|_{L^2}$.
 U is an ~~unitary~~ ^{isometry} operation.

Step 1: If $\varphi = \psi - U\psi$, then true. $(\star)$

$$\text{Pf.} \quad \frac{1}{n} \sum_{k=0}^{n-1} U^k \varphi = \frac{1}{n} \sum_{k=0}^{n-1} (U^k \psi - U^{k+1} \psi) = \frac{1}{n} (\psi - U^n \psi) \rightarrow 0$$

and, $\varphi \in \mathcal{H}^\perp$. so $P\varphi = 0$

Step 2: $\mathcal{L} = \{ \psi - U\psi, \psi \in L^2 \}$, then $\overline{\mathcal{L}} = \mathcal{H}^\perp$.

Pf. $\mathcal{L} \subset \mathcal{H}^\perp$. then $\overline{\mathcal{L}} \subset \mathcal{H}^\perp$.

If $\overline{\mathcal{L}} \subsetneq \mathcal{H}^\perp$. $\exists 0 \neq \varphi \in \mathcal{H}^\perp \cap \overline{\mathcal{L}}^\perp$,

But The $\forall \psi \in L^2$ $\langle \varphi, \psi - U\psi \rangle = 0$, U^* adjoint op

$$\Rightarrow \langle \varphi - U^* \varphi, \psi \rangle = 0 \Rightarrow \varphi = U^* \varphi,$$

(consider $\|U\varphi - \varphi\|^2 = \|\varphi\|^2 - 2\langle \varphi, U\varphi \rangle + \|U\varphi\|^2 = 2\|\varphi\|^2 - 2\langle U^* \varphi, \varphi \rangle$)

Hence $U\varphi = \varphi \Rightarrow \varphi \in \mathcal{H}$. Contradiction $\mathcal{H}^\perp \cap \mathcal{H} = \{0\}$.